

EXPERIMENT NO. : 2 (ASSIGNMENT)

DATE: 20/08/2026

1. AIM

To design and verify the schematic and layout of a Dynamic CMOS 2 input nand using 180 nm technology, and compare its pre-layout and post-layout simulation results

2. TOOL

Cadence Virtuoso

3. THEORY

A 2-input Dynamic CMOS NAND gate operates in two distinct phases: precharge and evaluation, controlled by a clock signal.

During the precharge phase, when the clock is LOW, the precharge PMOS transistor is ON and charges the output node to V_{dd} . The pull-down network (PDN) is disabled during this phase, so the output is precharged to logic HIGH.

During the evaluation phase, when the clock becomes HIGH, the precharge transistor is turned OFF and the PDN is enabled. The PDN consists of two NMOS transistors connected in series and controlled by inputs A and B . The output discharges to LOW only when both A and B are HIGH. For any other input combination, the output retains its precharged HIGH state.

The logic function of the 2-input Dynamic CMOS NAND gate is $Y = A \cdot B$. Since the output is already precharged to HIGH before evaluation, no LOW-to-HIGH transition occurs during evaluation. Therefore, $t_{pLH} = 0$ for the evaluation operation, while the relevant delay is t_{pHL} during discharge through the PDN.

The schematic of the 2-input Dynamic CMOS NAND gate is designed and simulated using 180 nm CMOS technology in Cadence Virtuoso. The corresponding physical layout is designed and verified using design-rule checking (DRC) and layout-versus-schematic (LVS) verification.

Pre-layout and post-layout simulations are then performed and compared to analyze the effect of layout parasitics on propagation delay and overall circuit performance.

4. TABLES

Table 4.1: CMOS Inverter Design Parameters

Parameter	Value	Unit
Technology	180	nm
V_{dd}	1.8	V
PMOS Width	400	nm
NMOS Width	400	nm

Table 4.2: Result comparision (pre layout vs post layout)

Parameter	Pre layout	Post layout	Unit	Error(wiht respect to pre layout)(%)
V_{th}	718.126	718.126	mV	0.00%
V_{OH}	1.8	1.8	V	0.00%
V_{OL}	0	0	V	0.00%
V_{IH}	846.93	847.26	mV	0.039%
V_{IL}	557.951	557.949	mV	0.0004%
NM_H	0.9531	0.9527	V	0.042%
NM_L	557.951	557.949	V	0.0004%
t_{HL}	34.8	50.9	ps	46.2644%
t_{LH}	0	0	ps	0.00%
t_p	34.8	50.9	ps	46.2644%
t_r	-49542.82	157.12	ps	100.3171%
t_f	41.7	68.3	ps	63.789%

Table 4.3: Result comparision (static vs dynamic for pre layout)

Parameter	Static	Dynamic	Unit
V_{th}	743.914	718.126	mV
V_{OH}	1.8	1.8	V
V_{OL}	0	0	V
V_{IH}	861.5719	846.33	mV
V_{IL}	565.54	557.95	mV
NM_H	0.9384	0.9537	V
NM_L	0.5655	0.5579	V
t_{pLH}	0.06168	0	ns
t_{pHL}	0.0687	0.043	ns
t_p	0.0652	0.0215	ns
t_r	0.099908	0.153	ns
t_f	0.037584	0.042	ns

Table 4.4: Result comparison (static vs dynamic for post layout)

Parameter	Static	Dynamic	Unit
V_{th}	715.843	718.126	mV
V_{OH}	1.8	1.8	V
V_{OL}	0	0	V
V_{IH}	836.9349	847.26	mV
V_{IL}	555.4868	557.949	mV
NM_H	0.9631	0.9527	V
NM_L	0.5555	0.5579	V
t_{pLH}	0.070606	0	ns
t_{pHL}	0.05478	0.1049	ns
t_p	0.0627	0.0524	ns
t_r	0.135	0.0862	ns
t_f	0.066719	0.1494	ns

5. CALCULATION

$$N_{MH} = V_{OH} - V_{IH} = 1.8 - 0.836654 = 0.963346V \approx 0.9633V$$

$$N_{ML} = V_{IL} - V_{OL} = 0.555495 - 0 = 0.555495V$$

$$N_{MH(post)} = V_{OH} - V_{IH} = 1.8 - 0.8369212 = 0.9630788V \approx 0.9631V$$

$$N_{ML(post)} = V_{IL} - V_{OL} = 0.5554912 - 0 = 0.5554912V$$

$$t_{HL(pre)} = (37.0219 - 37.0075) * 1000 = 14.4ps$$

$$t_{HL(post)} = (37.0302 - 37.0075) * 1000 = 22.7ps$$

$$t_{LH(pre)} = 0ps$$

$$t_{LH(post)} = 0ps$$

$$t_{p(pre)} = 37.0219ps$$

$$t_{p(post)} = 37.0302ps$$

$$t_{r(pre)} = (20.0648 - 20.0185) * 1000 = 46.3ps$$

$$t_{r(post)} = (20.1034 - 20.0234) * 1000 = 80.0ps$$

$$t_{f(pre)} = (37.0376 - 37.0137) * 1000 = 23.9ps$$

$$t_{f(post)} = (37.056 - 37.0154) * 1000 = 40.6ps$$

$$Error_{V_{th}} = (0.0004/715.711) * 100 = 0.0001$$

$$Error_{V_{IH}} = (0.2672/836.654) * 100 = 0.0319$$

$$Error_{V_{IL}} = (0.0038/555.495) * 100 = 0.0007$$

$$Error_{N_{MH}} = (0.0002/0.9633) * 100 = 0.0208$$

$$Error_{N_{ML}} = (0.0000038/0.555495) * 100 = 0.0007$$

$$Error_{t_p} = (0.0083/37.0219) * 100 = 0.0224$$

CIRCUIT DIAGRAMS & WAVEFORMS

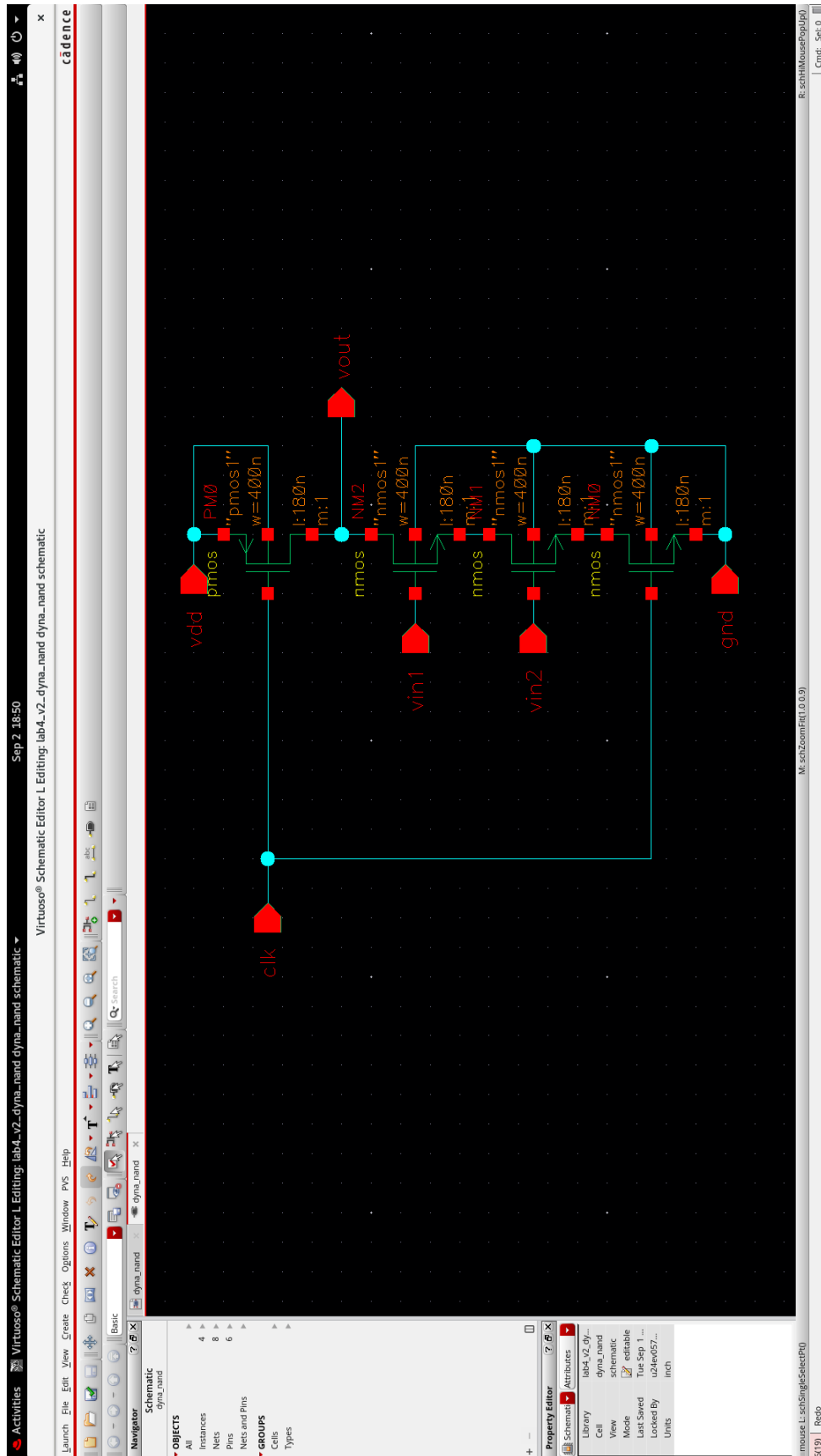


Figure .1: Schematic

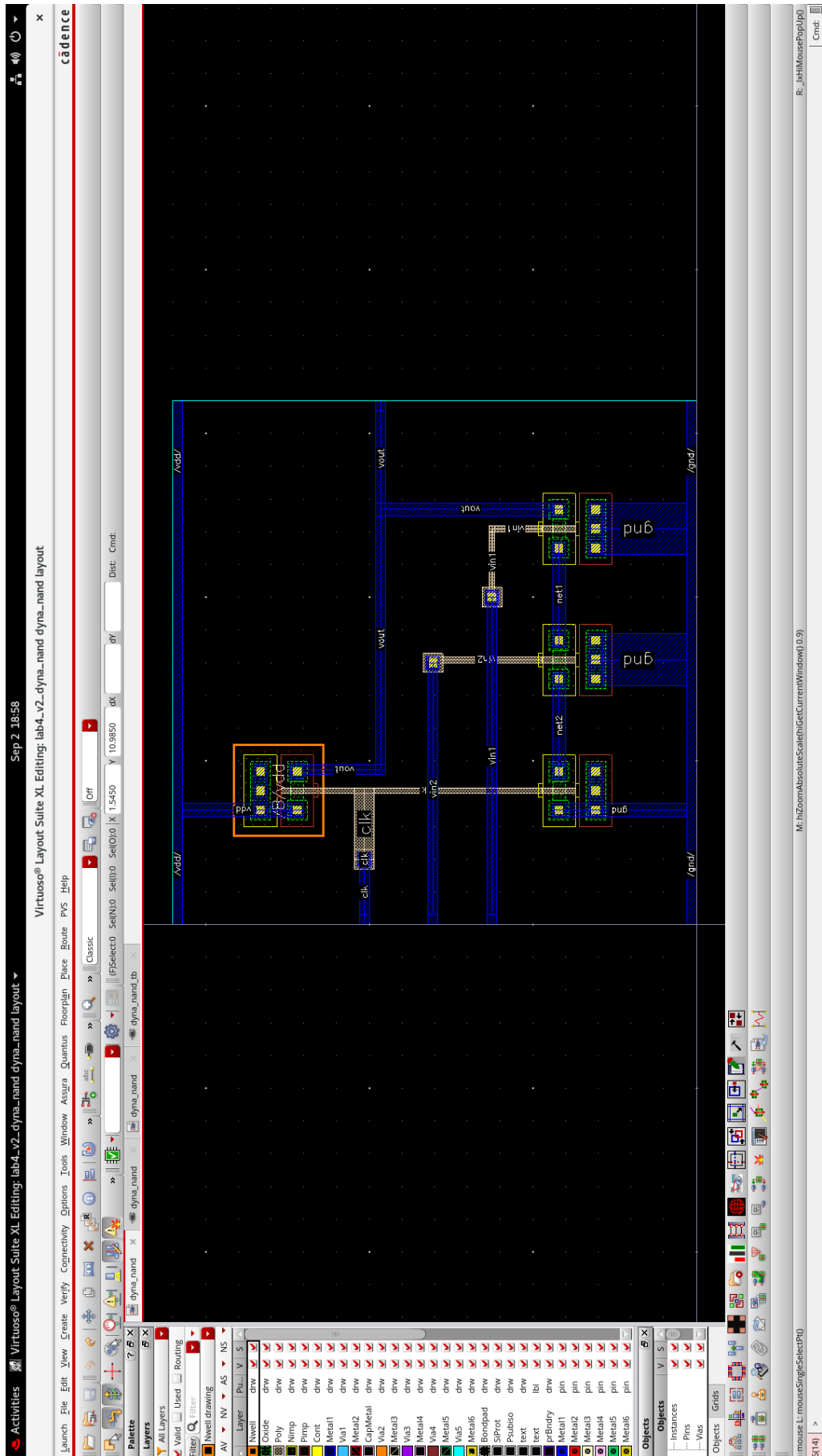


Figure .2: Layout

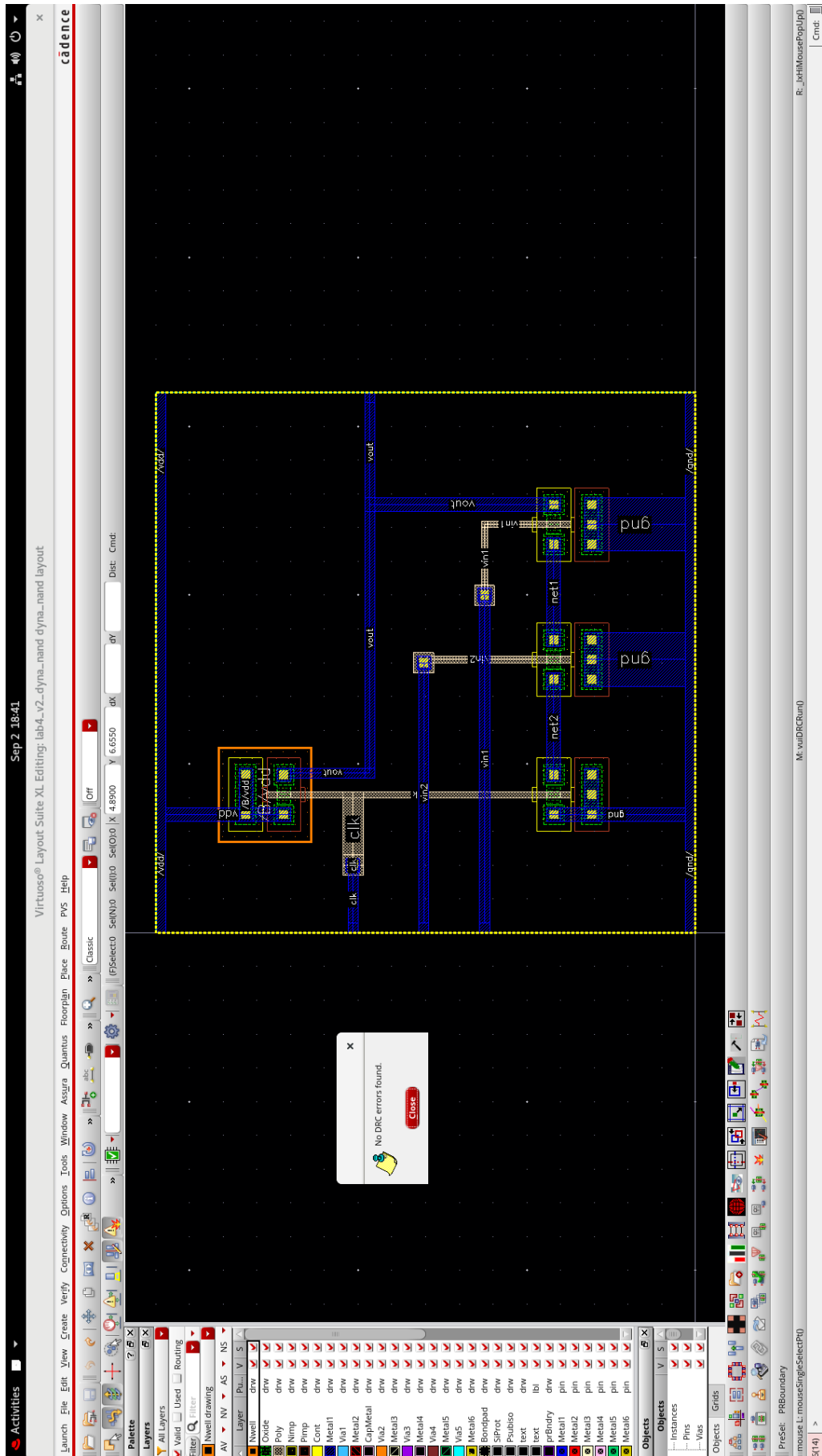


Figure .3: Drc

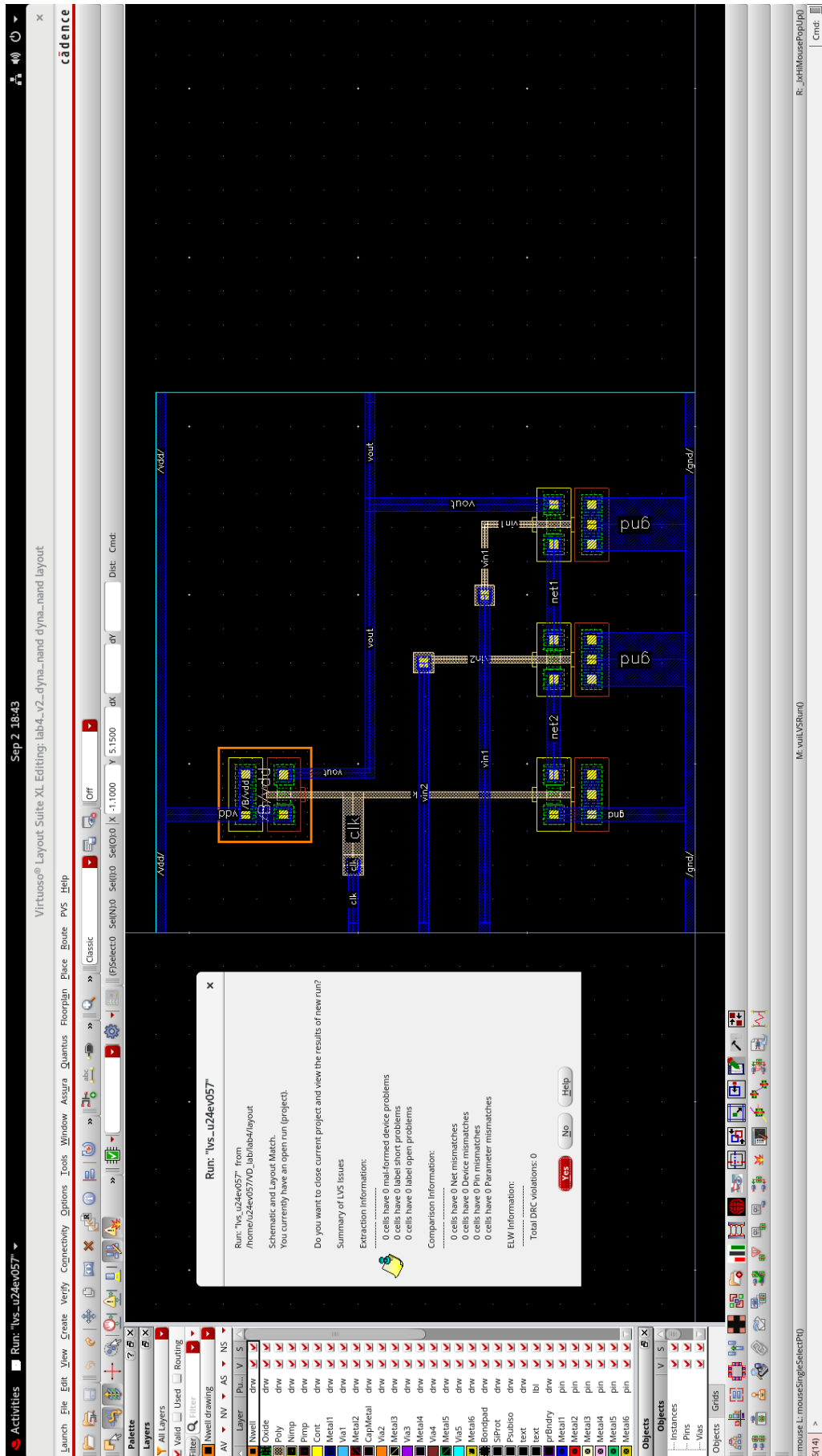


Figure .4: Lvs

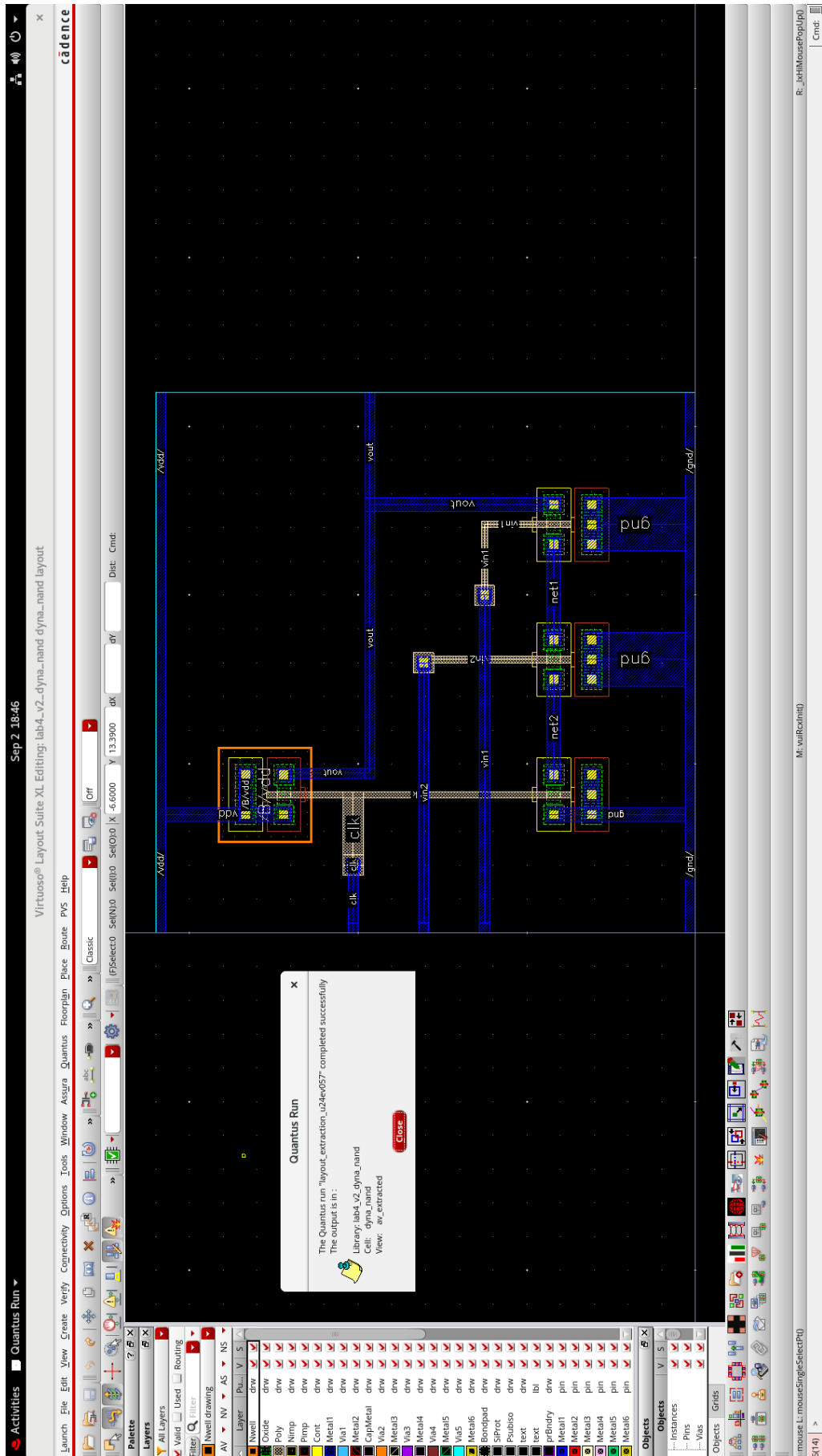


Figure .5: Quantus

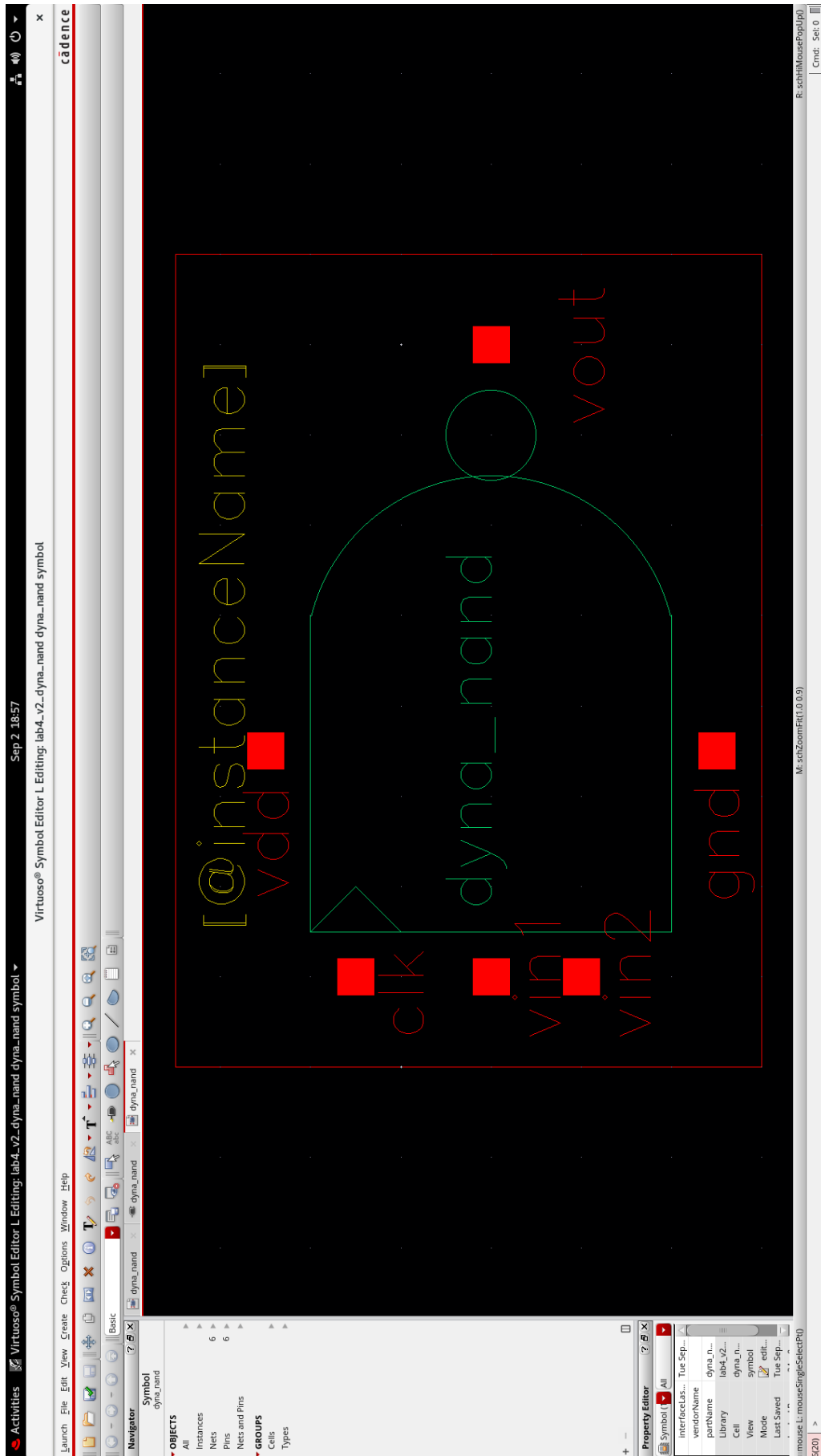


Figure .6: Symbol

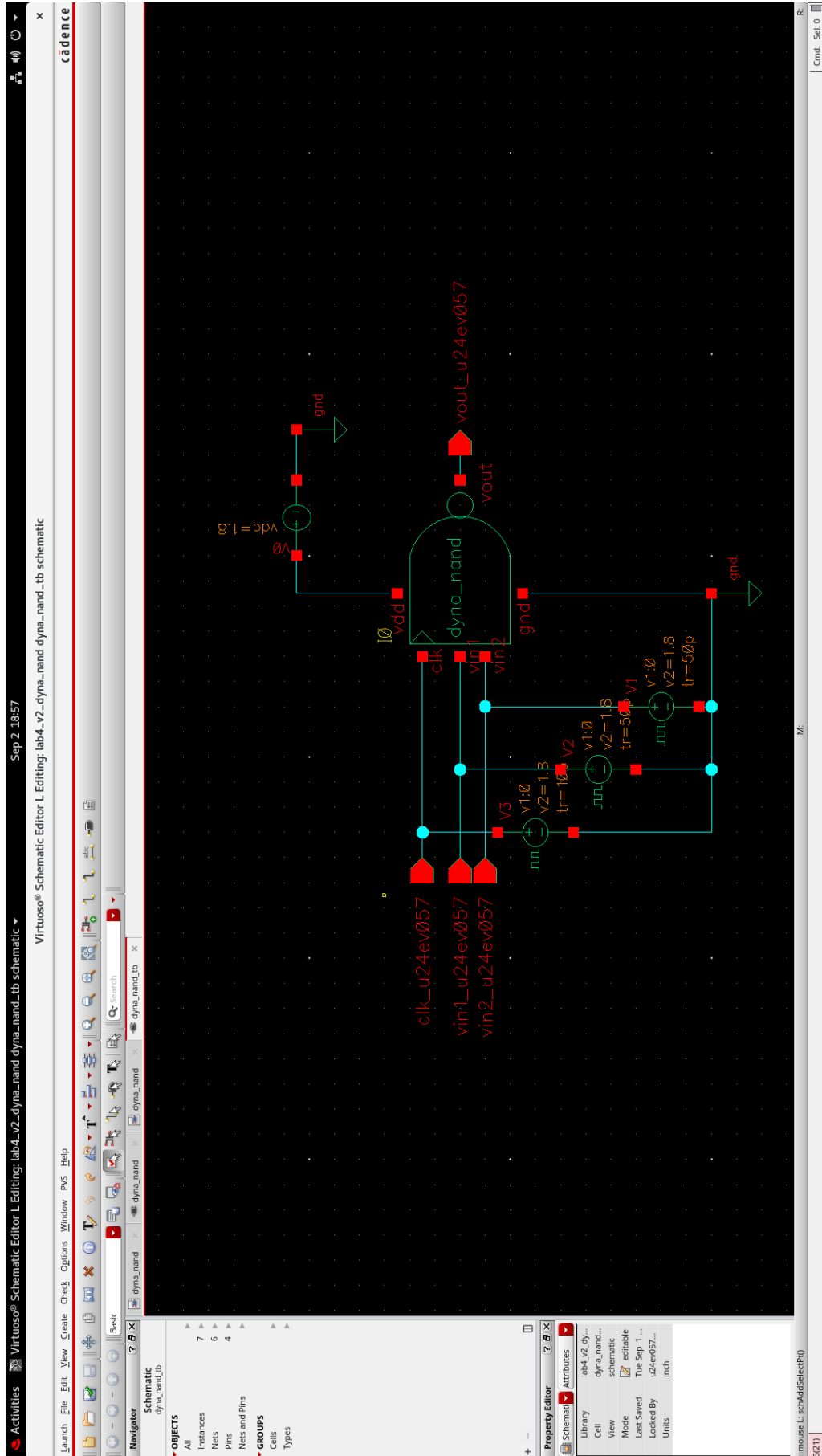


Figure .7: Tb

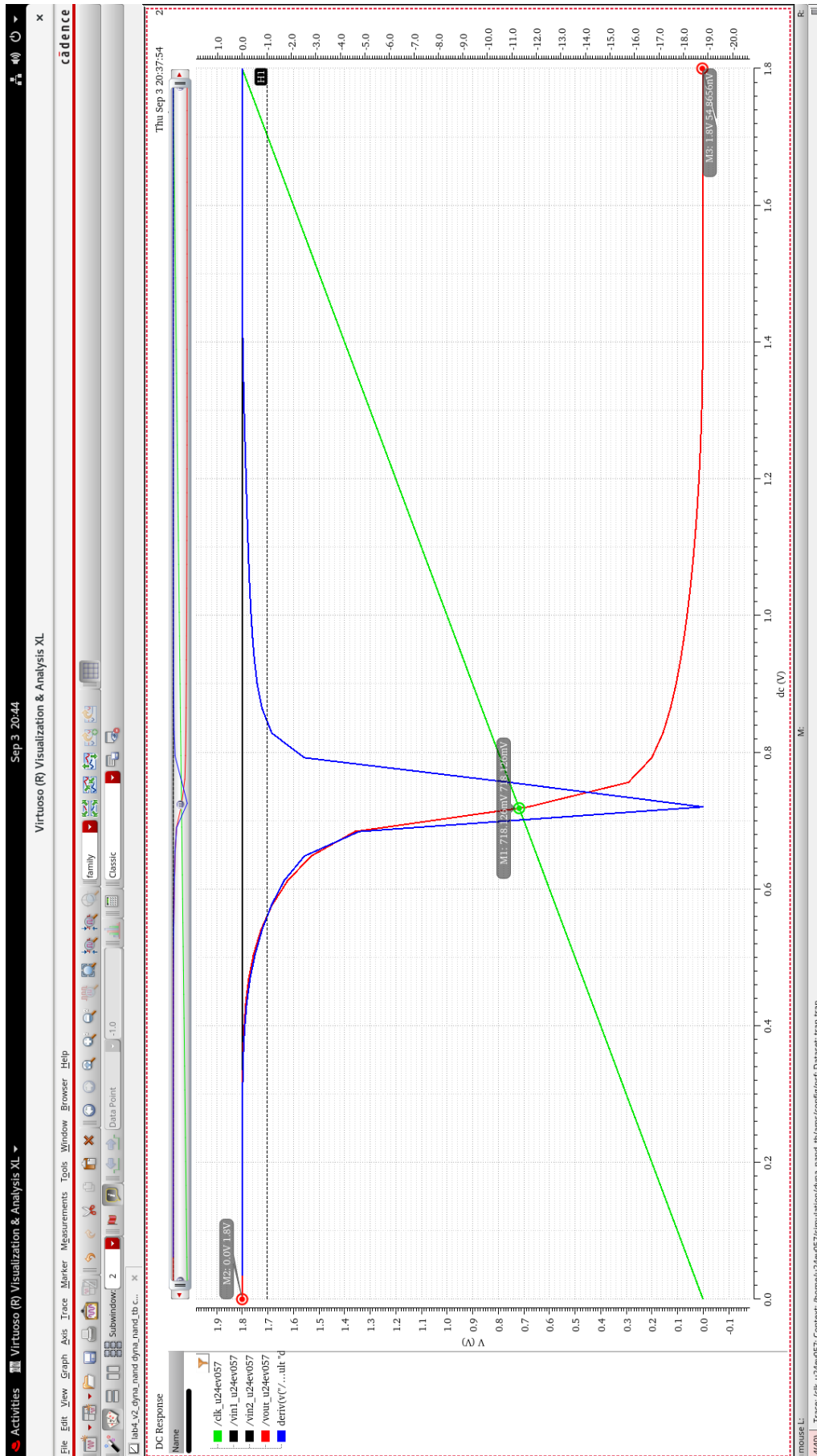


Figure .8: Dc Pre Layout

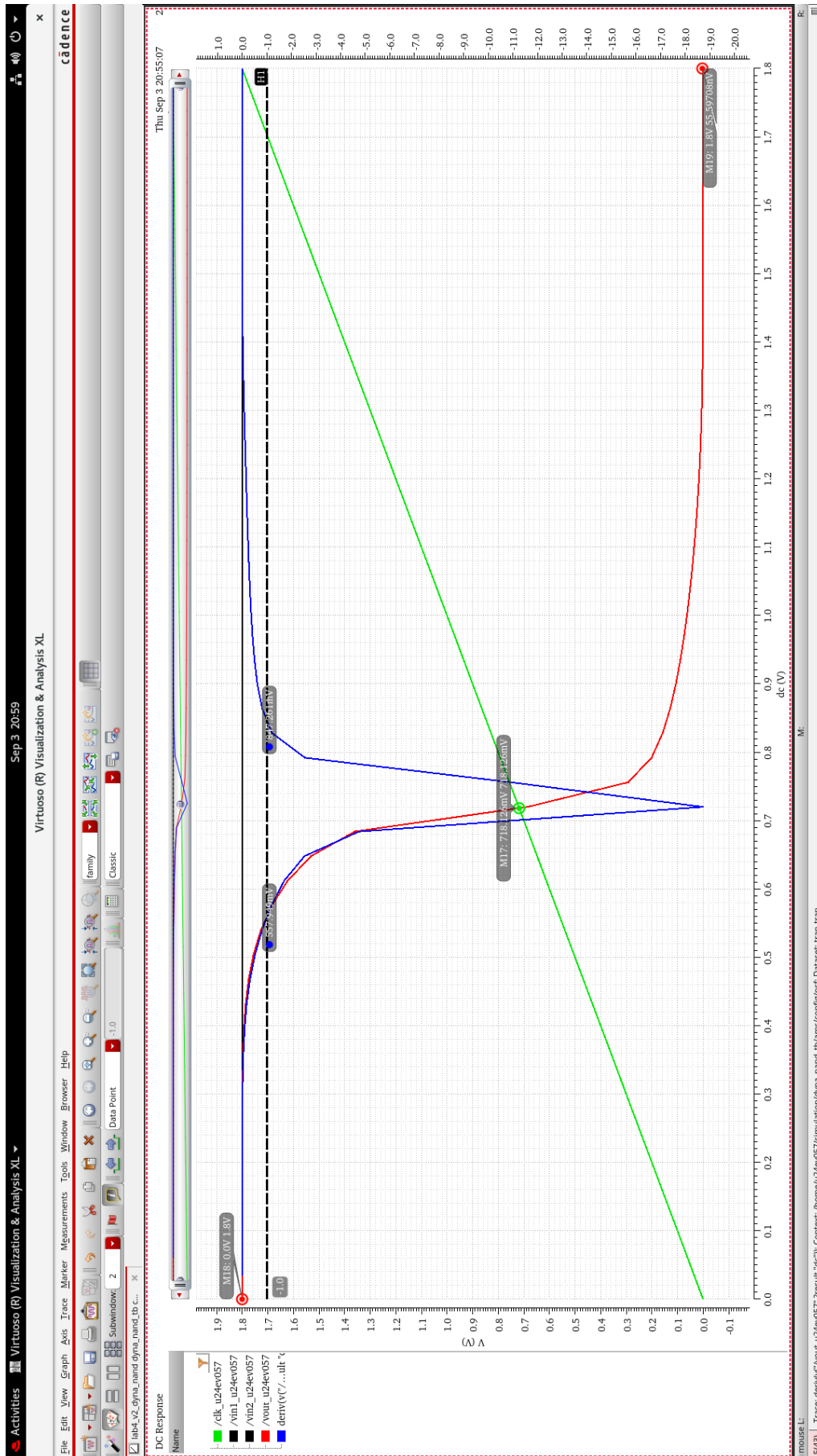


Figure .9: Dc Post Layout

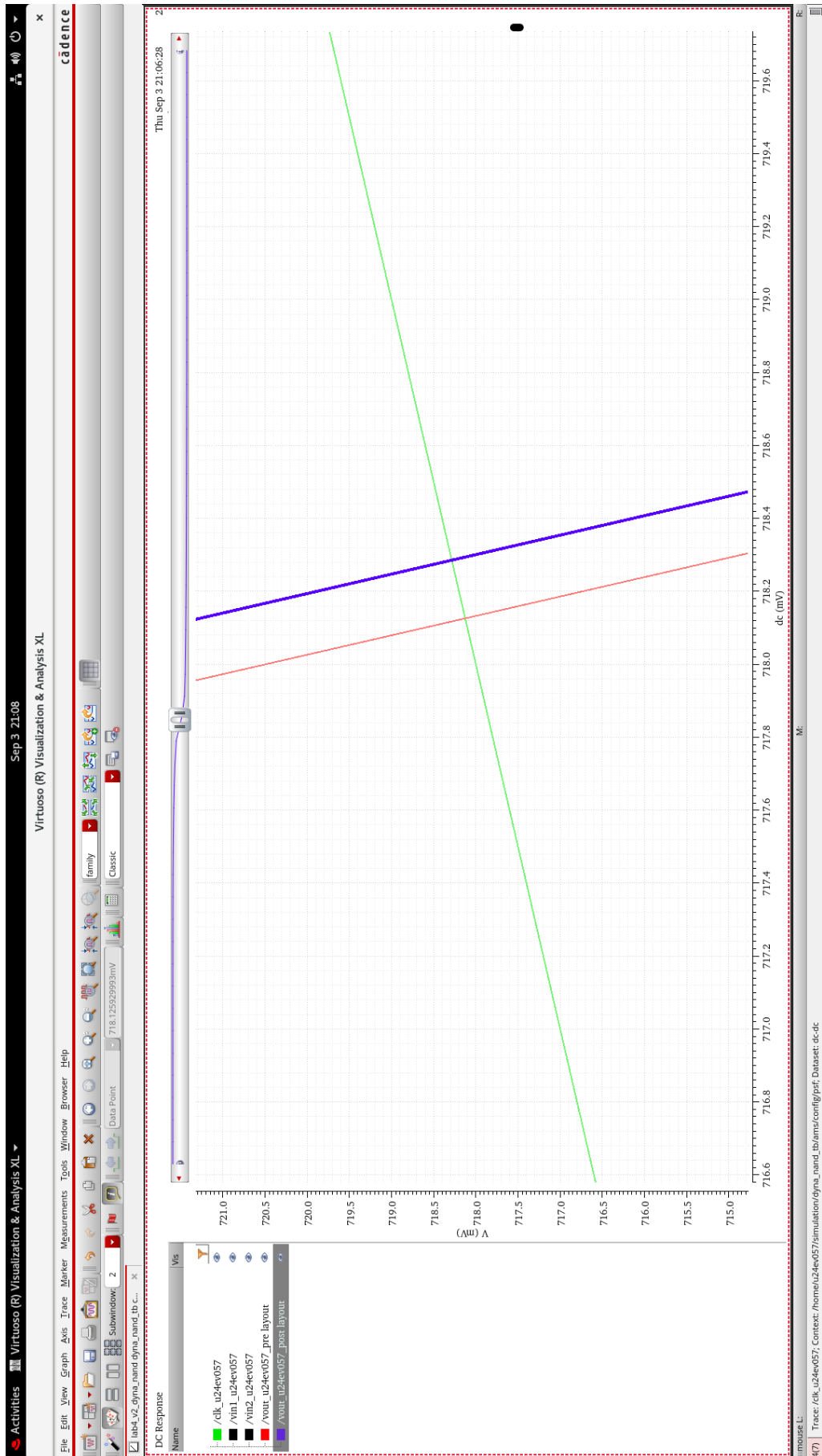


Figure .10: Dc Pre Vs Post

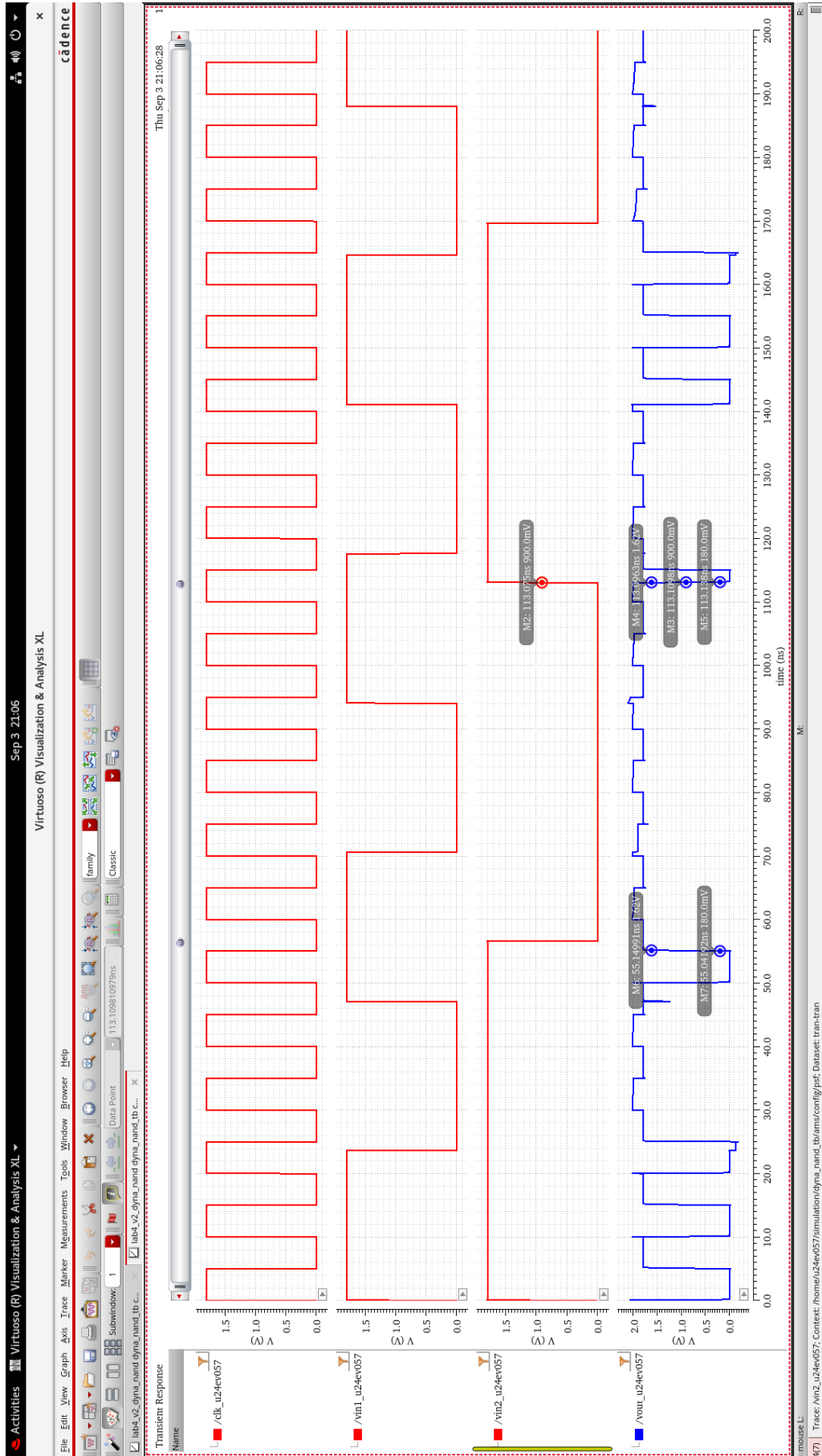


Figure .11: Tran Pre Layout

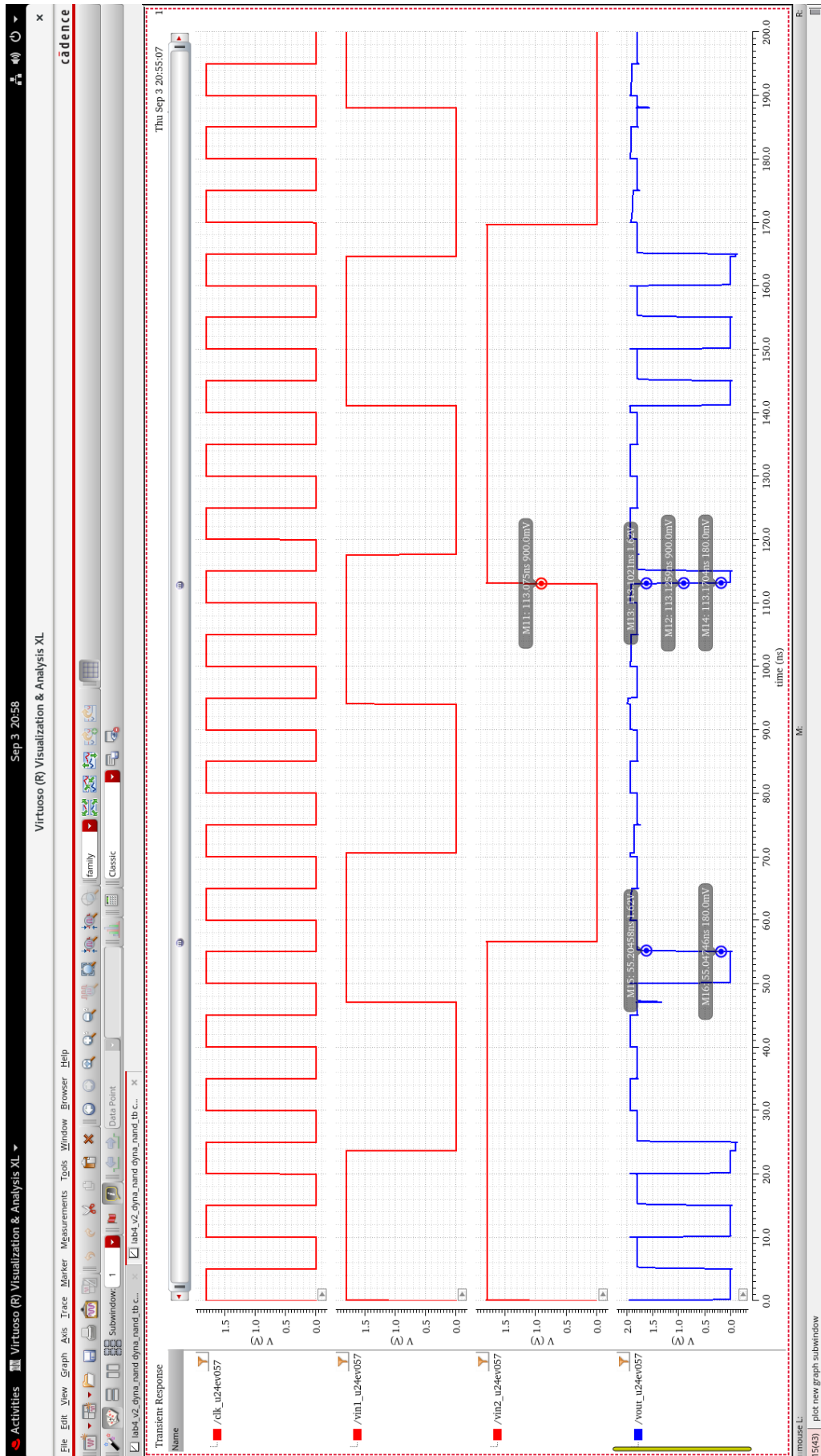


Figure .12: Tran Post Layout

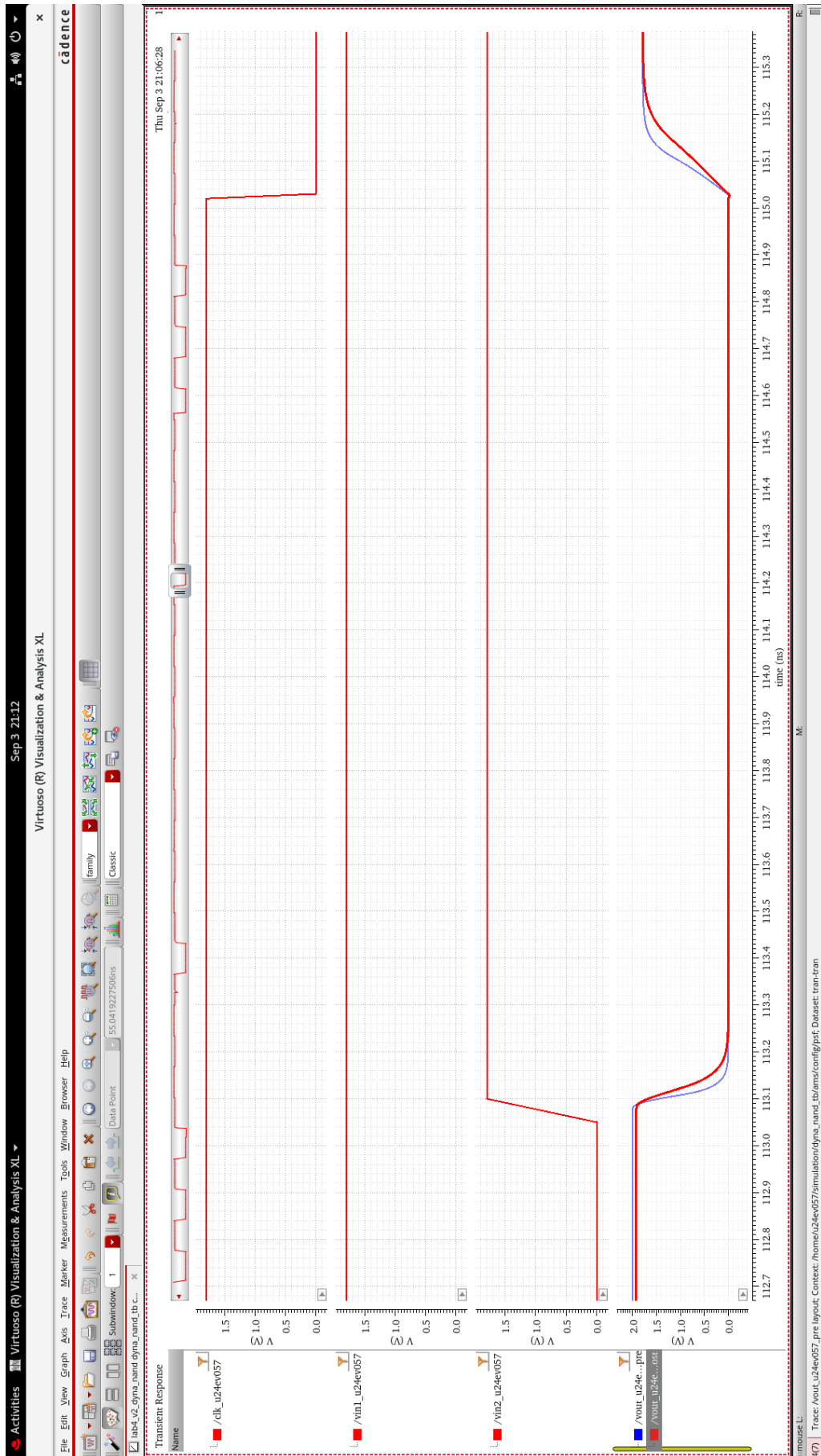


Figure .13: Tran Pre Vs Post

7. RESULT

The 2-input Dynamic CMOS NAND gate was successfully designed and verified using 180 nm CMOS technology, and the pre-layout and post-layout simulation results were compared to evaluate the effect of layout parasitics on circuit performance.